Youngjae (Jimmy) Sim

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EDUCATION

University of California, Berkeley — Berkeley, CA

B.A. Applied Mathematics, B.A. Computer Science — Aug. 2026 – May 2029

- **Relevant Coursework:** Multivariable Calculus, Discrete Mathematics, Structure and Interpretation of Computer Programs, Introduction to Quantitative Finance, Poker Theory & Fundamentals
- **Relevant Involvement:** Berkeley Math Tournament (Problem Writer), Berkeley Trading Invitational 2026, Citadel Securities Trading Challenge, Jane Street Guts++ Competition, Poker@Berkeley, Real Estate Club, Capital Investments at Berkeley (Quant Education)

Korea International School — Seongnam, South Korea

High School Diploma (GPA: 3.95/4.0, SAT: 1560) — Aug. 2022 – May 2026

RESEARCH & TECHNICAL EXPERIENCE

Presolve for Conic Quadratic Optimization — Berkeley, CA

Undergraduate Contributor | Rust (Mentored by Matteo Santamaria, UC Berkeley PhD Student) — Sep. 2026 – Present

- Contribute to a Rust library implementing presolve reductions for conic quadratic programs, aimed at reducing problem size and computational cost before solving.
- Construct and test optimization examples to trace fixed-variable elimination, conversion of constraints into variable bounds, infeasibility detection, and solution recovery.
- Develop mdBook documentation explaining the mathematical reasoning and implementation of presolve reduction rules.

Stock Price Prediction: Comparative Deep Learning Study — Remote

Independent Project | Python, TensorFlow/Keras — May 2025 – Sep. 2025

- Designed and benchmarked RNN, LSTM, and a custom conditional VAE forecaster on 5 years of stock data, cutting RMSE by up to 62% over a vanilla RNN through gated memory and a hand-built KL-regularized training loop.

Science Internship Program (SIP), UC Santa Cruz — Santa Cruz, CA

Research Intern (selected, 1 of 3) — Jun. 2024 – Aug. 2024

- Built a Haskell API for secure distributed applications, using monadic design for fault tolerance and deadlock-free data flow.
- Presented final research to 60+ faculty, peer researchers, and parents.

Anti-Blur Pothole Detection and Embedded Board Implementation Research — Remote

Independent Researcher | Published, Journal of Student Research, Vol. 14 (2025) — Apr. 2024 – Oct. 2024

- Trained and evaluated U-Net/CNN object detection models in Python to improve detection robustness on blurred, real-world images, and deployed the model on an NVIDIA Jetson board for real-time inference.

ML-Based Cardiovascular Disease Detection via Retinal Imaging Research — Remote

Independent Researcher | Published, Journal of Student Research, Vol. 13 (2024) — Jun. 2023 – Dec. 2023

- Developed age-guided CNN classification models using retinal imaging biomarkers in YCbCr color space to automate CVD diagnosis.

LEADERSHIP & ACTIVITIES

Competitive Programming & Computer Programming Clubs – President — Aug. 2022 – May 2026

- Coached 40+ members for international contests; organized the International School Programming Cup (60+ participants, 5 schools).
- Directed SoriHack, a school-wise hackathon fusing generative AI and music composition, from concept through judging and prizes.

Quiz Bowl (National Academic Quiz Tournament) – Team Captain — Oct. 2023 – May 2026

- Led team to a KAIAC (nationals) title and the HS Quiz Bowl Asian Championships (Singapore, 2025).

Phoenix Talks Club – Founder & President — Dec. 2023 – May 2026

- Built a speaker series connecting students with industry leaders and academics; hosted an MIT student panel (70+ attendees)

Varsity Cross-Country & Track and Field – Team Captain — Aug. 2022 – May 2026

- Captained both programs to podium finishes in national and international meets; broke two school relay records.

AWARDS & HONORS

Gold Division, USA Computing Olympiad (2024) · **3rd Place**, Korea Youth Investment Competition (2024) · **Gold Medal**, Korea Science & Engineering Fair (2024) · **Gold Medal**, International Greenwich Olympiad (2024) · **Selected Presenter**, IEEE ISEC, Princeton University (2025) · **Semifinalist**, Columbia Junior Science Journal (2024-25) · **Commended Poet**, Foyle Young Poets of the Year Award (2025) · **Published Author**, Blue Marble Review (2026) · **Selected Participant**, Iowa Young Writers' Studio, University of Iowa (2025)

SKILLS & INTERESTS

Languages: Python, C++, Rust, Haskell, R, TensorFlow/Keras | **Quantitative:** Statistical modeling, game theory, market making
Interests: Quiz bowl, Film/Cinema, Poker, Reading (classics, philosophy, nonfiction), Prediction Markets